

### Question 14

1 mark

There are 2 types of pencils, 3 colours of highlighters, and 5 styles of pens.

If you must select one of each to form a set, how many different sets of writing instruments are possible?

- a) 10
- b) 11
- c) 25
- d) 30

### Question 15

1 mark

The point  $P(\theta)$  lies on the unit circle. What are the coordinates of the point  $P$  if  $\theta = 120^\circ$ ?

- a)  $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b)  $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$
- c)  $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$
- d)  $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

### Question 16

1 mark

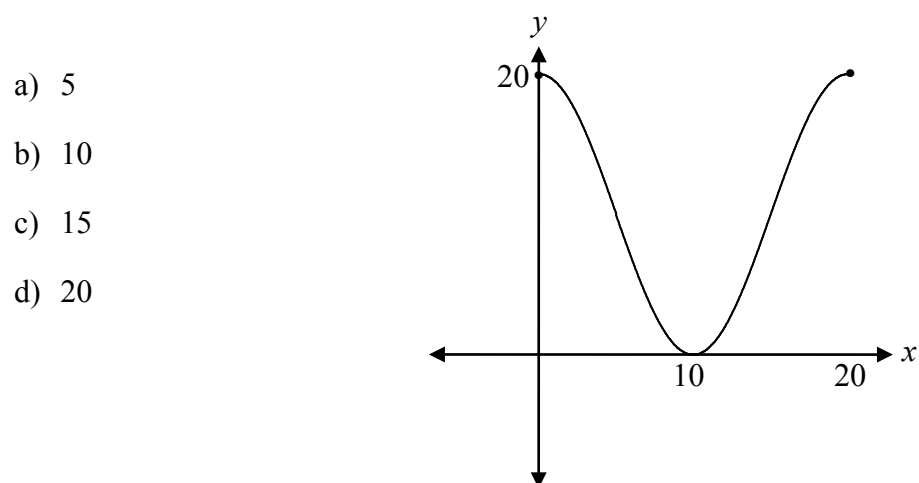
Identify the function that has a domain of  $\{x|x \geq 7\}$  and a range of  $\{y|y \geq 0\}$ .

- a)  $f(x) = \sqrt{x} + 7$
- b)  $f(x) = \sqrt{x} - 7$
- c)  $f(x) = \sqrt{x+7}$
- d)  $f(x) = \sqrt{x-7}$

### Question 17

1 mark

Using  $y = -10 \cos[B(x - C)] + D$ , the value of  $C$  that corresponds to the following graph is:



### Question 18

1 mark

Given the following row of Pascal's Triangle, identify the binomial expansion with these coefficients.

1      5      10      10      5      1

a)  $(x + y)^4$

b)  $(x + y)^5$

c)  $(x + y)^6$

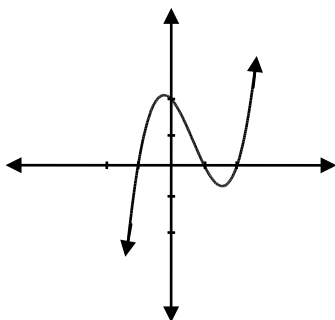
d)  $(x + y)^7$

### Question 19

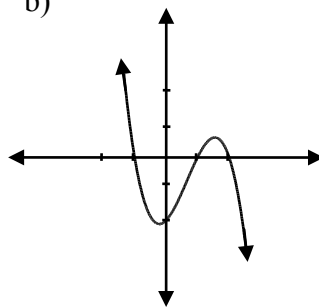
1 mark

Identify the graph of the function  $f(x) = -(x - 2)(x - 1)^2(x + 1)$ .

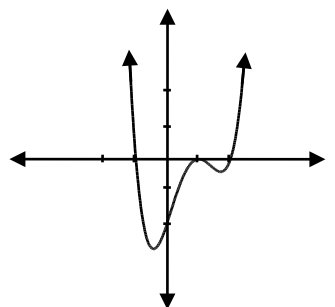
a)



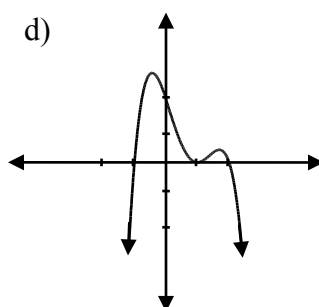
b)



c)



d)



### Question 20

1 mark

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Determine the value of  $\log_9(\log_3 27)$ .

- a)  $\frac{1}{3}$
- b)  $\frac{1}{2}$
- c) 2
- d) 3

### Question 21

1 mark

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If  $(x, y)$  is a point on the graph of  $y = f(x)$ , identify the coordinates of this point on the graph of  $g(x) = f(2x) + 5$ .

- a)  $\left(\frac{x}{2}, y + 5\right)$
- b)  $(2x, y + 5)$
- c)  $\left(\frac{x}{2}, y - 5\right)$
- d)  $\left(\frac{x}{2} - 5, y\right)$

## Question 22

1 mark

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Identify an equivalent expression for  $1 + \log_2 5$ .

- a)  $\log_2 5$
- b)  $\log_2 7$
- c)  $\log_2 10$
- d)  $\log_2 11$

### Question 23

4 marks

114

Solve:

$$2\log_4 x - \log_4 (x + 3) = 1$$

## Question 24

3 marks 115

The following transformations are applied to  $f(x)$ , resulting in a new function,  $g(x)$ .

- reflection over the  $y$ -axis
- horizontal translation of 3 units to the right
- vertical translation of 4 units down

Write the equation of  $g(x)$  in terms of  $f(x)$ .

$g(x) =$  \_\_\_\_\_

### Question 25

a) 3 marks   b) 1 mark

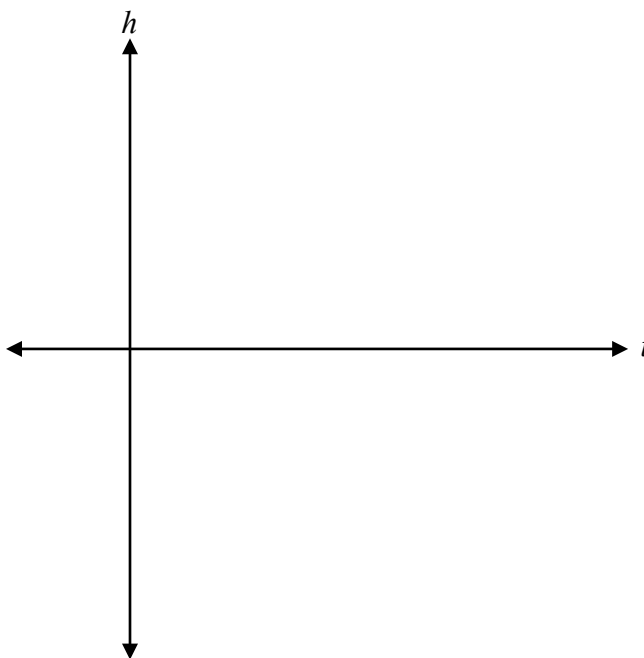
116  
117

The height of a bicycle pedal as the bicycle is moving at a constant speed can be represented by the following function:

$$h(t) = 15 \cos \frac{2\pi}{5}t + 30$$

where  $h$  is the height of the pedal above the ground, in cm, and  $t$  is the time, in seconds.

- a) Sketch a graph of at least one period of this function, where  $t \geq 0$ .



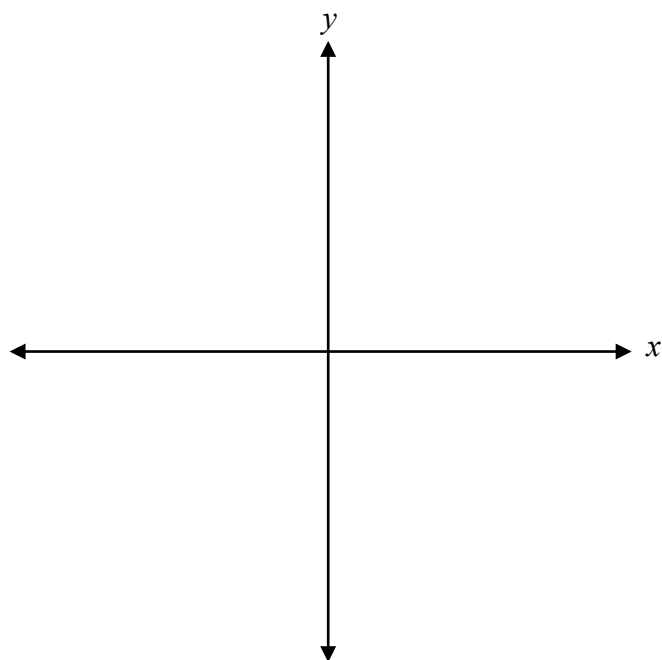
- b) Determine the height of the bicycle pedal at 7.5 seconds.

### Question 26

4 marks 118

Sketch the graph of the function  $f(x)$  and determine the  $y$ -intercept.

$$f(x) = \frac{4}{(x-2)(x+2)}$$



$y$ -intercept: \_\_\_\_\_

## Question 27

1 mark 119

Kim solved the following logarithmic equation:

$$\log_2 \left( -\frac{x}{3} \right) = \log_2 (x - 4)$$

$$-\frac{x}{3} = x - 4$$

$$-x = 3x - 12$$

$$-4x = -12$$

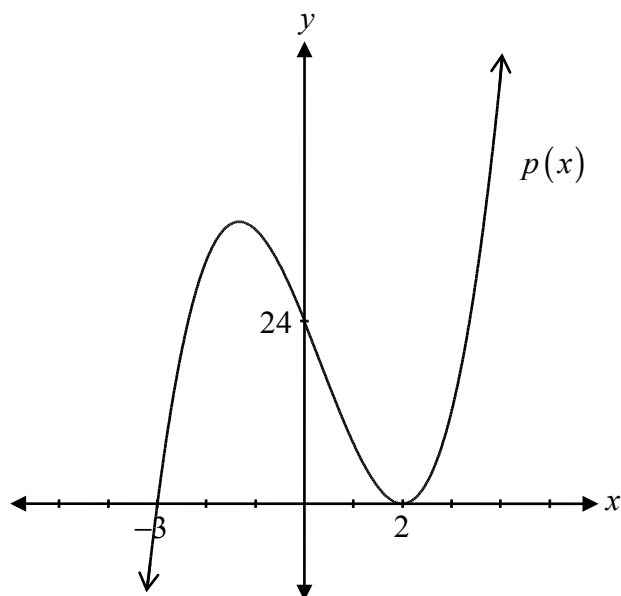
$$\cancel{x = 3}$$

Explain why  $x = 3$  is an extraneous solution.

# Question 28

3 marks 120

Determine the equation of the polynomial function represented by the graph.



$p(x) =$  \_\_\_\_\_

### Question 29

1 mark 121

Determine the coterminal angles with  $\frac{2\pi}{3}$  over the interval  $[-2\pi, 4\pi]$ .

### Question 30

a) 1 mark   b) 1 mark

122  
123

- a) Solve the following equation:

$$0 = \sqrt{4x - 8} - 2$$

- b) Explain how your answer in a) is related to the graph of  $y = \sqrt{4x - 8} - 2$ .

### Question 31

3 marks 124

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Determine the exact value of  $\sin \frac{13\pi}{12}$ .

### Question 32

a) 2 marks    b) 1 mark

125  
126

Given the functions  $f(x) = x + 2$  and  $g(x) = \frac{1}{x-5}$ :

- a) Determine the equation of the composite function  $f(g(x))$  and its domain.

$$f(g(x)) = \underline{\hspace{2cm}}$$

domain:  $\underline{\hspace{2cm}}$

- b) Determine the  $x$ -intercept and  $y$ -intercept of  $f(g(x))$ .

$x$ -intercept:  $\underline{\hspace{2cm}}$

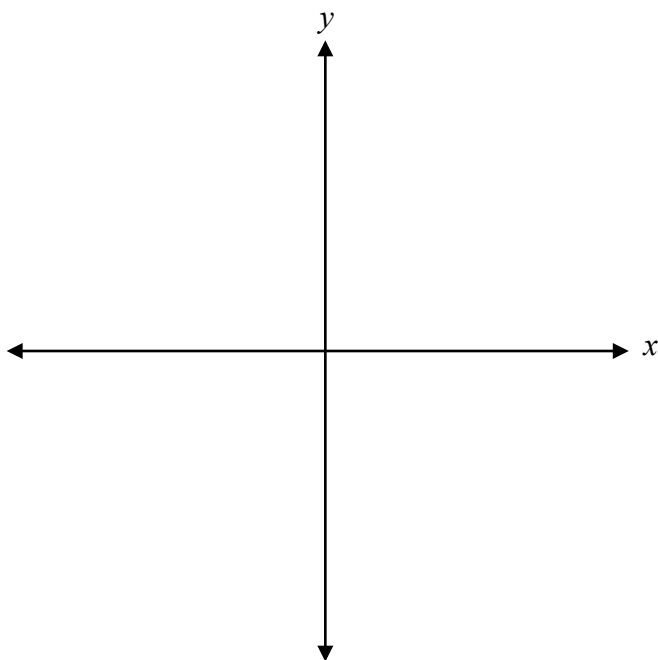
$y$ -intercept:  $\underline{\hspace{2cm}}$

### Question 33

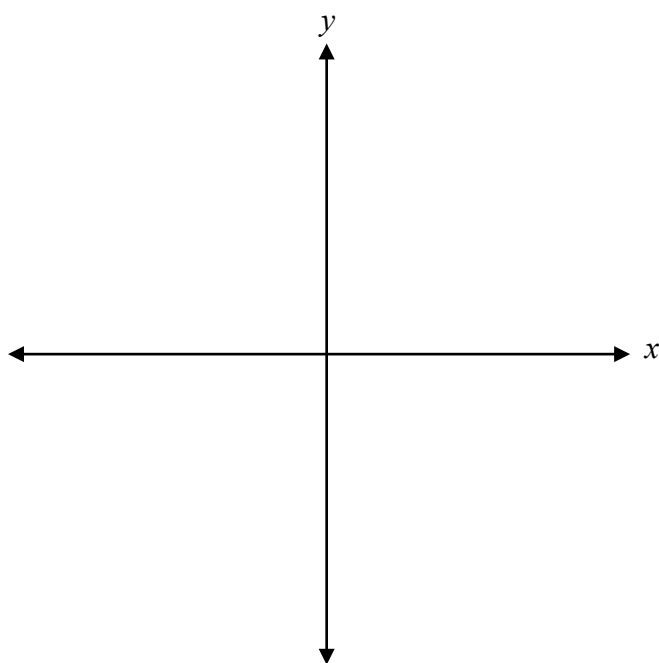
a) 2 marks    b) 1 mark

127  
128

- a) Sketch the graph of  $f(x) = \log_5(x-1)$ .



- b) Sketch the graph of  $f^{-1}(x)$ .



### Question 34

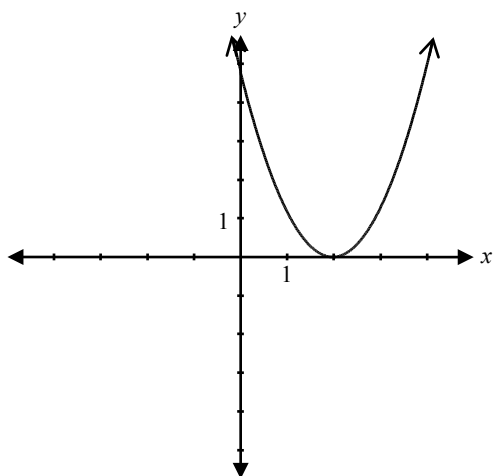
1 mark 129

Explain what the graph of a rational function looks like near a vertical asymptote.

### Question 35

1 mark 130

Given the graph of  $f(x) = (x - 2)^2$ ,



determine one possible restriction for the domain of  $f(x)$  so that its inverse is a function.

Domain: \_\_\_\_\_

### Question 36

2 marks 131

Over the interval  $[0, 2\pi]$ , determine the non-permissible values of  $\theta$  in the expression  $\csc \theta (\cos \theta + 1)$ .

### Question 37

1 mark <sup>132</sup>

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Explain why  ${}_3C_8$  is undefined.

### Question 38

3 marks 133

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Solve:

$${}_nP_3 = 48(n-1)$$

### Question 39

a) 2 marks   b) 1 mark   c) 1 mark

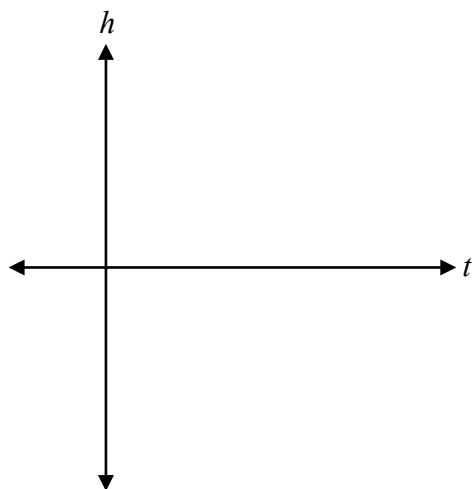
134  
135  
136

Christine dives off a diving board.

Her dive is modelled by the function  $h(t) = t^3 - 3t^2 - t + 3$ , where  $h$  is her height in metres, relative to the water surface and  $t$  is the time in seconds after diving off the diving board.

a) Given that  $(t + 1)$  is a factor for the function  $h(t)$ , determine the other factors.

b) Sketch the graph of the function  $h(t)$  for the time interval  $t = 0$  to  $t = 3$ .



c) Determine how many seconds Christine is underwater.

## Question 40

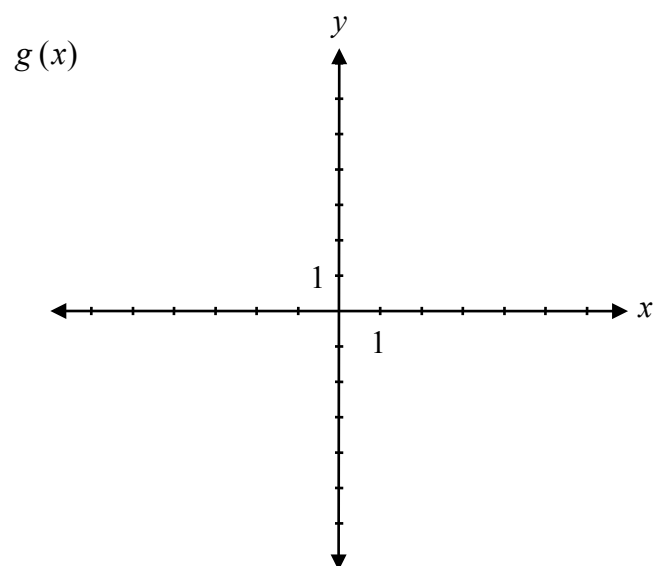
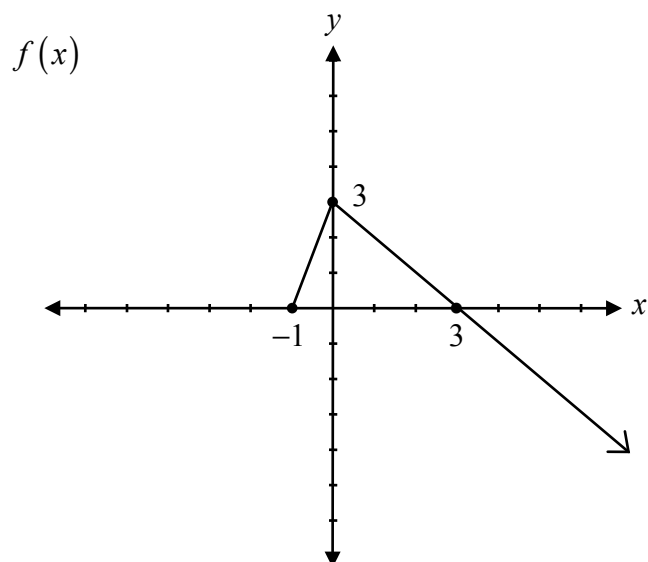
3 marks 137

Prove the identity for all permissible values of  $x$ .

$$\sec x + \tan x = \frac{\cos x}{1 - \sin x}$$

| Left-Hand Side | Right-Hand Side |
|----------------|-----------------|
|                |                 |

Given the graph of  $f(x)$ , sketch the graph of the function  $g(x) = -|f(x)|$ .



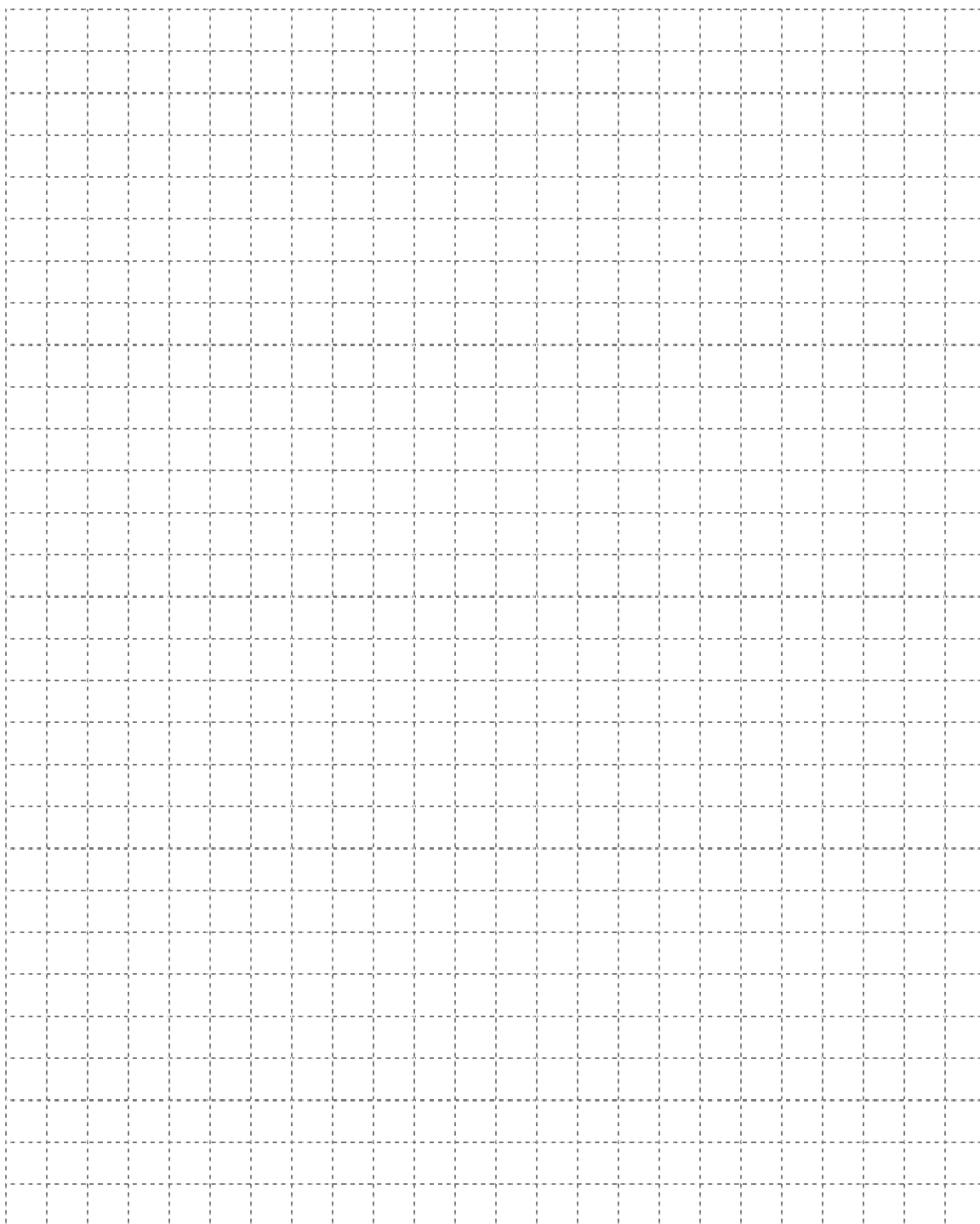
### Question 42

2 marks 139

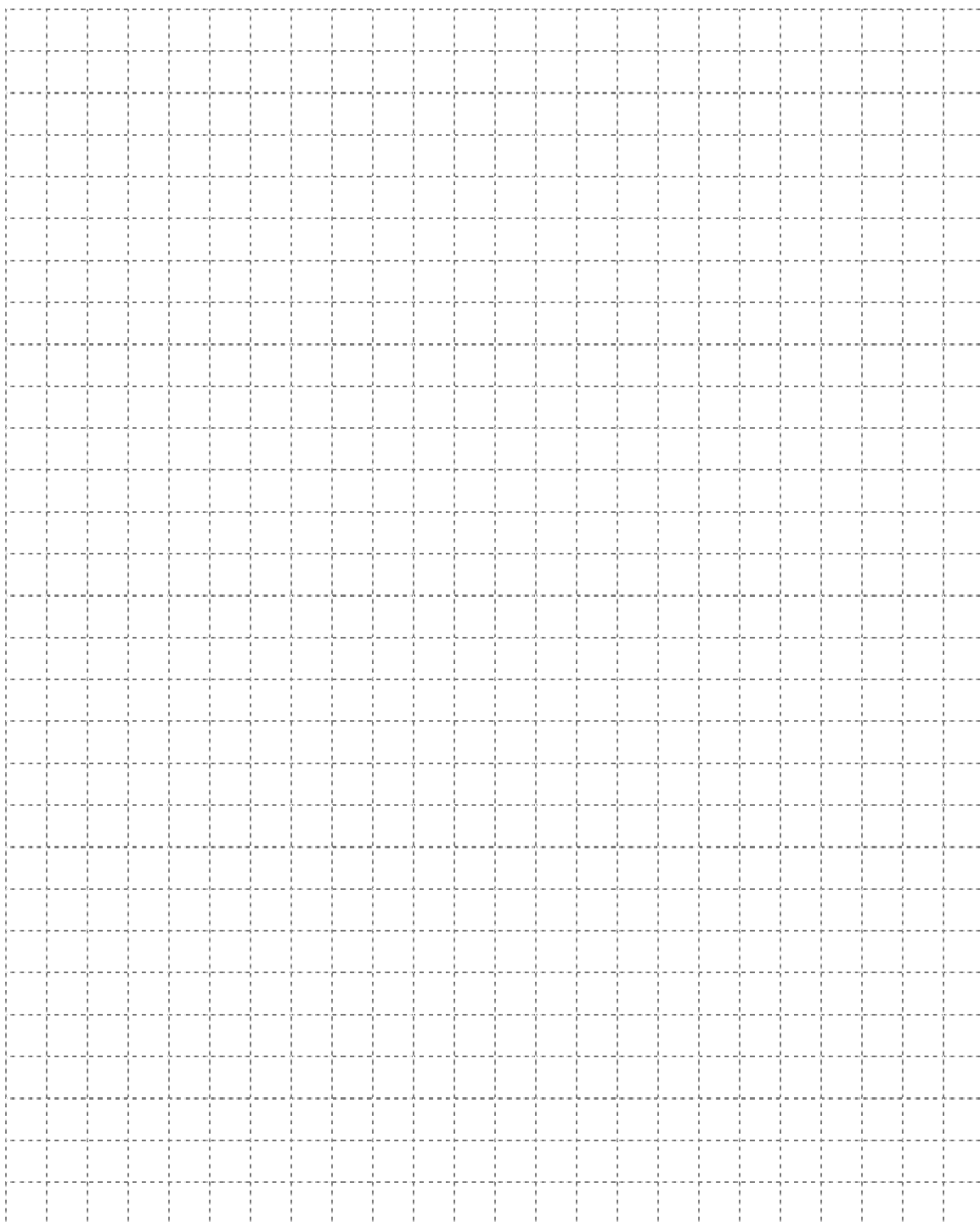
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Given that  $\cot \theta = -\frac{2}{5}$ , where  $\theta$  is in Quadrant IV, determine the exact value of  $\sin \theta$ .

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